

Software Reliability (Nishu)

- Bathub curve of hardware reliability.
- Software reliability curve (failure rate versus time).
- Software may be retired only if it becomes obsolete.

Some of the contributing factors are given below: --

- * change in environment
- * change in infrastructure/technology
- * major change in requirements
- * increase in complexity
- * extremely difficult to maintain
- * deterioration in structure of the code
- * slow execution speed
- * poor graphical user interface

Definition of Software Reliability:

- As per IEEE standard, "Software reliability is defined as the ability of a system or component to perform its required functions under stated conditions for a specified period of time."

- software reliability ~~is~~: It is the probability of a failure free operation of a program for a specified time in a specified environment.

Failures and Faults

- Failure is dynamic: The term failure relates to the behaviour of the program
- A fault is the defect in the program that, when executed under particular conditions causes a failure
- A fault is a property of the program that, when executed under particular conditions causes a failure
- A fault is a property of the program rather than a property of its execution or behaviour.

=> Three kinds of time :-

- ① Execution time for a program is the time that is actually spent by a processor in executing the instructions of that program.
- ② calendar time is the familiar time that we normally experience.
- ③ clock time represents the elapsed time from start to end of program execution on a running computer.

=> There are four general ways of characterising failure occurrences in time :

- ① time of failure
- ② time interval between failures
- ③ cumulative failures experienced upto a given time
- ④ failures experienced in a time interval

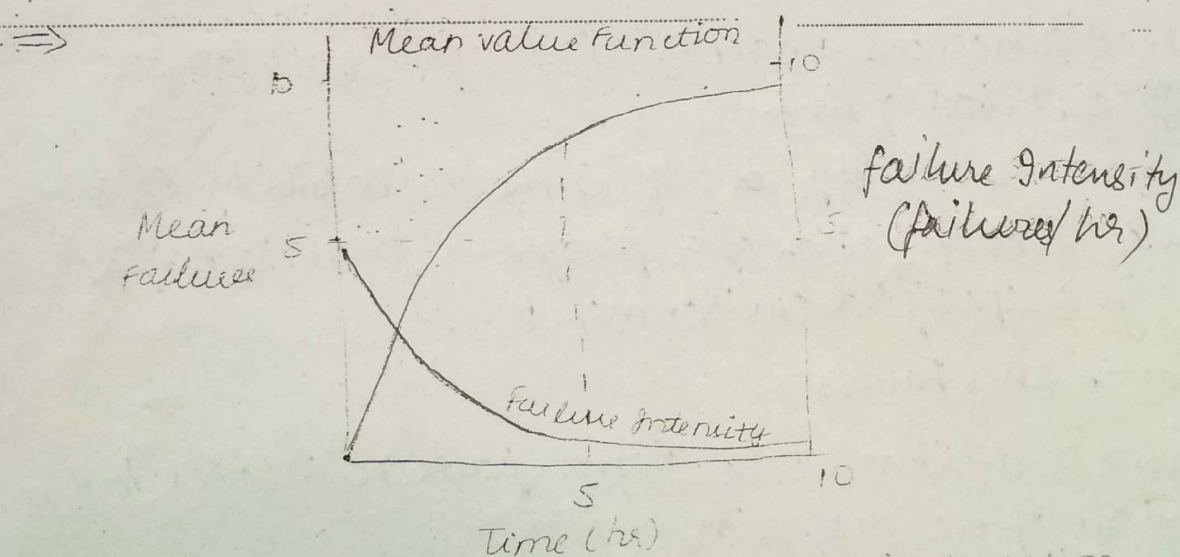


Fig: Mean value and Failure intensity functions

- The mean value function represents the average cumulative failures associated with each time point.
- The failure intensity function is the rate of change of the mean value function or the number of failures per unit time.
- Failure behaviour is affected by two principal factors:
 - ① The number of faults in the s/w being executed.
 - ② The execution environment & the operational profile of execution.

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- ① S/w reliability measures are used to evaluate software engineering technology quantitatively.
- ② S/w reliability measures offer you the possibility of evaluating development status during the test phases of a project.
- ③ S/w reliability measures are used to monitor the operational performance of software and to control new features added and design changes made to the S/w.
- ④ It enables us to ~~do~~ have a quantitative understanding of software quality and the various factors influencing it.

SOFTWARE RELIABILITY MODELS

→ Three principal factors that affect S/W reliability :-

- ① Fault introduction
- ② Fault removal
- ③ Environment

→ Fault introduction: depends primarily on the characteristics of the developed code (code created or modified for the application) and development process characteristic, which include software engg technologies and tools used and level of experience of personnel.

→ Fault removal depends upon time, operational profile and the quality of repair activity.

→ The environment directly depends on the operational profile

→ Software reliability models are generally formulated in terms of the random processes.

→ Reliability models ^{that} are based on execution time yield better results than those based on calendar time.

→ Execution time will be denoted by τ and
'calendar time' " " t .

→ A stochastic process whose probability distribution ^{expe} changes over time is called non-homogeneous

$\lambda(t)$ = failure intensity function (Average number of failures per unit of time at time t)

BASIC EXECUTION TIME MODEL

✓ The model was developed by J.D MUSA in 1979 and is based on execution time

✓ → It is assumed that failures may occur according to a non-homogeneous Poisson Process (NHPP)

✓ → Real world events may be described using Poisson processes. Examples of Poisson processes are :-

* number of telephone calls expected in a given period of time

* expected number of road accidents in a given period of time

* expected number of persons visiting in a shopping mall in a given period of time.

→ In this case, processes are non-homogeneous since failure intensity changes as a function of time.

→ In this model, the decrease in failure intensity as a function of the number of failures observed is constant and is given as :

$$\lambda(\mu) = \lambda_0 \left(1 - \frac{\mu}{V_0} \right)$$

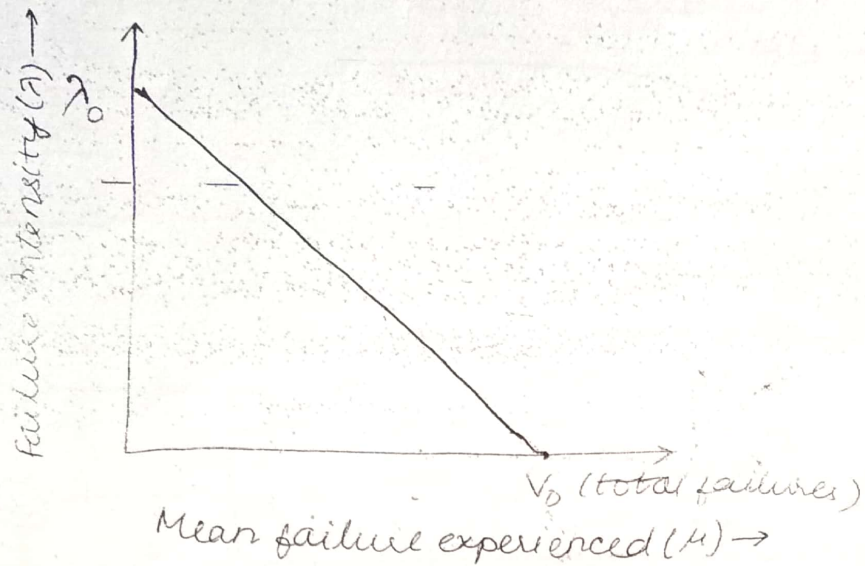
Decrease in failure intensity as no. of failure observed

where λ_0 : Initial failure intensity at the start of execution

V_0 : Number of failures experienced, if program is executed for infinite time period

μ : Average or expected number of failures experienced at a given point in time (mean failure experiences)

The relationship b/w failure intensity (λ) and mean failure experienced (μ) is given below :-



→ In this model, the decrease in failure intensity, as a function of the number of failures observed, is constant and is given as :

$$\lambda(\mu) = \lambda_0 \left(1 - \frac{\mu}{V_0} \right) \quad \text{————— ①}$$

where λ_0 : Initial failure intensity at the start of execution

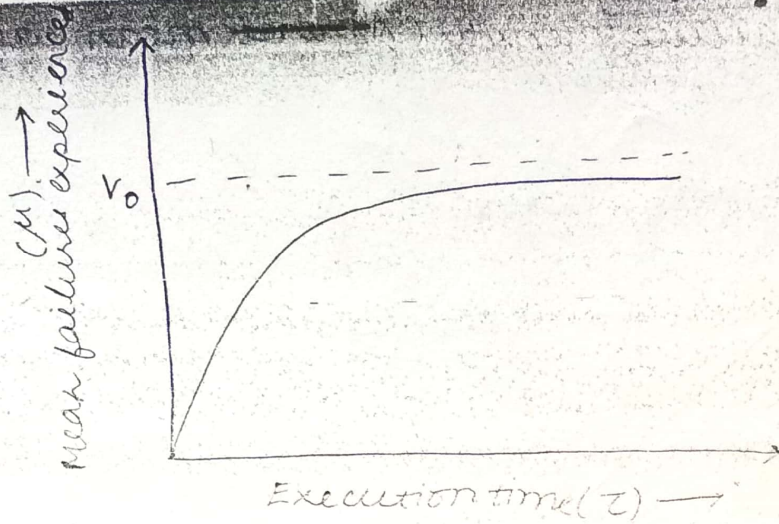
V_0 : Number of failures experienced, if program is executed for infinite time period

μ : Average or expected number of failures experienced at a given point in time

→ The slope of the failure intensity can be obtained by finding its first derivative and is given as :

$$\frac{d\lambda}{d\mu} = -\frac{\lambda_0}{V_0} \quad \text{————— ②}$$

The negative sign shows that there is a negative slope meaning thereby a decrementing trend in failure intensity.

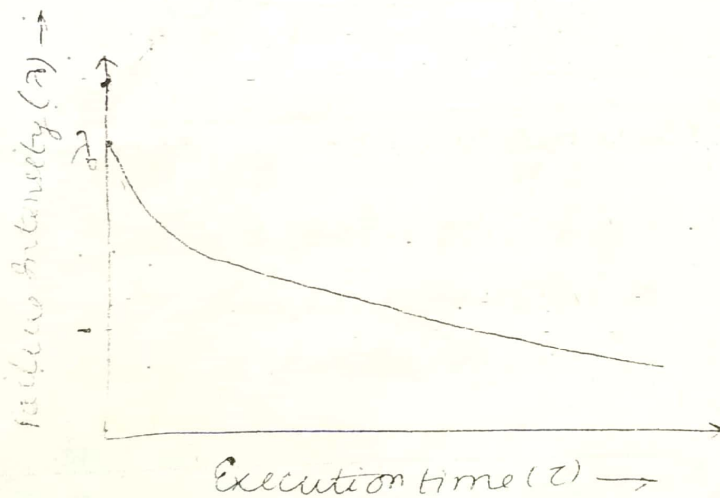


Relationship
b/w z & μ
for basic model

From eq ①,
$$\frac{d\mu(z)}{dz} = \lambda_0 \left(1 - \frac{\mu(z)}{V_0} \right) \quad \text{--- ③}$$

The above equation can be solved for $\mu(z)$,

$$\mu(z) = V_0 \left[1 - \exp\left(-\frac{\lambda_0 z}{V_0}\right) \right] \quad \text{--- ④}$$

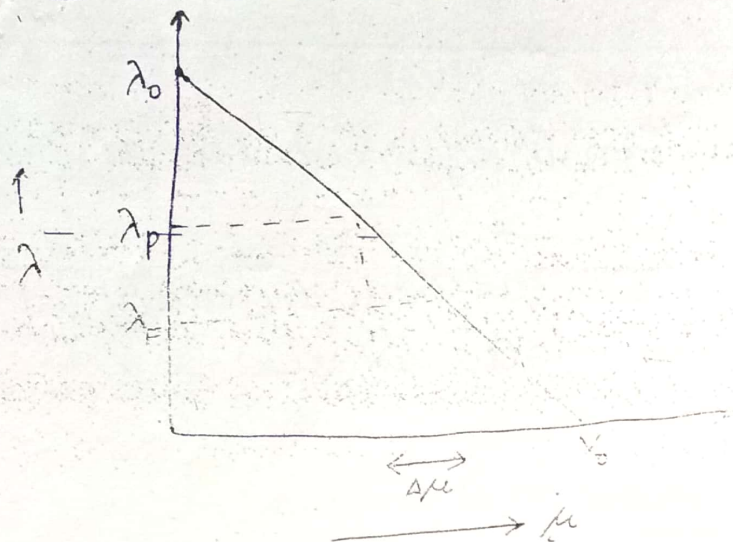


Failure intensity
versus execution time
for basic model

→ The relationship is useful for determining the present failure intensity at any given value of execution time. Differentiating eq ④ results in:

$$\lambda(z) = \lambda_0 \exp\left(-\frac{\lambda_0 z}{V_0}\right) \quad \text{--- ⑤}$$

Derived quantities



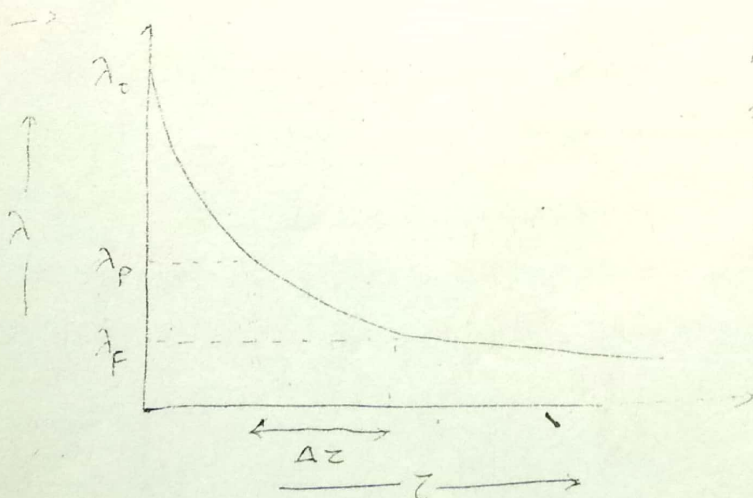
λ_0 : Initial failure intensity
 λ_p : Present failure intensity
 λ_F : Failure intensity Objective
 $\Delta\mu$: Expected number of additional failures to be experienced to reach failure intensity objective

Fig: Additional failures required to be experienced to reach the objective

→ Assume that we have chosen a failure intensity objective for the S/W product. Assume some failures have been corrected through removing the associated faults. Then, we can use the objective and present failure intensity to determine the additional failures that must be experienced to reach that objective.

→ $\Delta\mu$ can be derived in mathematical form as

$$\Delta\mu = \frac{V_0}{\lambda_0} (\lambda_p - \lambda_F)$$



Additional execution time Δz required to reach the failure intensity objective can be shown in mathematical form as

$$\Delta z = \frac{V_0}{\lambda_0} \ln \left(\frac{\lambda_p}{\lambda_F} \right)$$

Hence $\Delta\mu$ and Δz give us the idea about failure correction workload.

Fig Additional time required to reach the objective

LOGARITHMIC POISSON EXECUTION TIME MODEL.

- This model is also developed by MUSA. ✓
- The failure intensity function is different here as compared to Basic model.
- In this case, failure intensity function (decrement per failure) decreases exponentially whereas it is constant for basic model.
- The failure intensity function is given as :

$$\lambda(\mu) = \lambda_0 \exp(-\theta\mu)$$

where θ is called the failure intensity decay parameter.

- The relationship b/w failure intensity (λ) and mean failures experienced (μ) is shown in fig

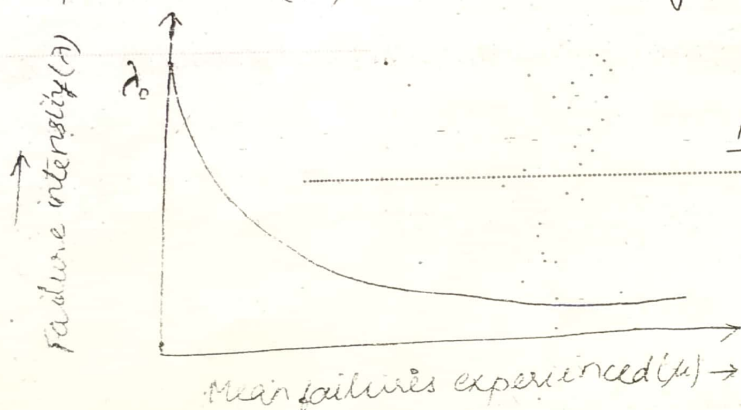


Fig: Relationship b/w μ & λ

- The θ represents the relative change of failure intensity per failure experienced. The slope of failure intensity function.

$$\frac{d\lambda}{d\mu} = -\lambda_0 \theta \exp(-\mu\theta)$$

$$\frac{d\lambda}{d\mu} = -\theta\lambda$$

The relationship b/w execution time and mean failures experienced is given in fig.

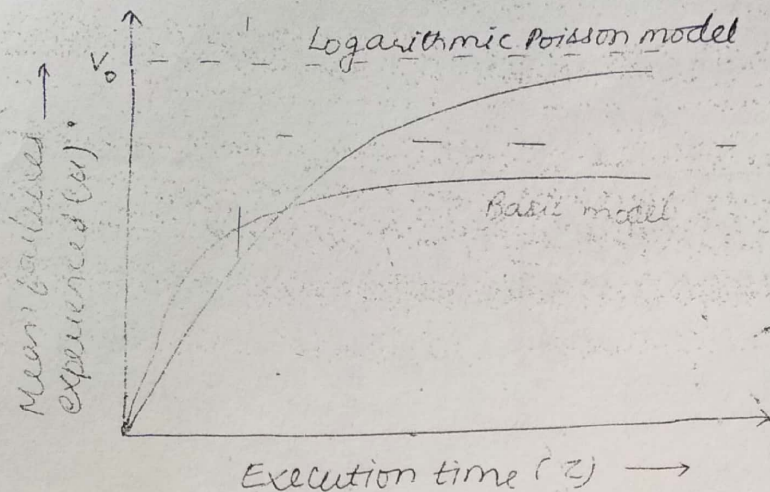


Fig: Relationship b/w z & μ

→ The expected number of failures for this model is always infinite at infinite time. The relation for number of failure is given by:

$$\mu(z) = \frac{1}{\theta} \ln(\lambda_0 \theta z + 1)$$

→ The expression for failure intensity is given as:

$$\lambda(z) = \lambda_0 / (\lambda_0 \theta z + 1)$$

Additional number of failures and additional execution time in this model are:

$$\Delta\mu = \frac{1}{\theta} \ln\left(\frac{\lambda_F}{\lambda_P}\right)$$

$$\Delta z = \frac{1}{\theta} \left[\frac{1}{\lambda_F} - \frac{1}{\lambda_P} \right]$$

where λ_P = Present failure intensity

λ_F = Failure intensity objective.

where

$\lambda_0 \rightarrow$ Initial failure intensity at start of execution.

$V_0 \rightarrow$ No. of failure experienced, if program is executed for finite time period

$\mu \rightarrow$ Average or expected no. of failure experienced at a given point in time or mean failure

$\tau \rightarrow$ execution time

$\lambda \rightarrow$ failure intensity

$\lambda_p \rightarrow$ present failure intensity

$\lambda_F \rightarrow$ Failure intensity objective

$\Delta\mu \rightarrow$ expected no. of additional failure to be experienced to reach failure intensity objective

Difference

BASIC

① Failure intensity function at failure observed

$$\lambda(u) = \lambda_0 \left(1 - \frac{\mu}{V_0} \right)$$

②

$$\mu(\tau) = V_0 \left(1 - \exp \left(-\frac{\lambda_0 \tau}{V_0} \right) \right)$$

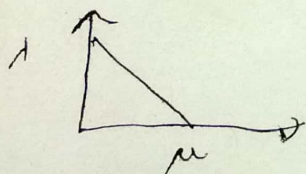
③ Failure intensity at execution time

$$\lambda(\tau) = \lambda_0 \exp \left(-\frac{\lambda_0 \tau}{V_0} \right)$$

④ $\Delta\mu = \frac{V_0}{\lambda_0} (\lambda_p - \lambda_F)$

⑤ $\Delta\tau = \frac{V_0}{\lambda_0} \ln \left(\frac{\lambda_p}{\lambda_F} \right)$

⑥ λ & μ



LOGARITHMIC

① Failure intensity function

$$\lambda(u) = \lambda_0 \exp(-\theta\mu)$$

$\theta \rightarrow$ failure intensity decay parameter

②

$$\mu(\tau) = \frac{1}{\theta} \ln(\lambda_0 \theta \tau + 1)$$

③

$$\lambda(\tau) = \lambda_0 / (\lambda_0 \theta \tau + 1)$$

④ $\Delta\mu = \frac{1}{\theta} \ln \left(\frac{\lambda_p}{\lambda_F} \right)$

⑤ $\Delta\tau = \frac{1}{\theta} \left[\frac{1}{\lambda_F} - \frac{1}{\lambda_p} \right]$

